

Database Description and Business Rules

AI-Assisted Pet Allergy Symptom Tracking System

Sam Cashion | 700619652 | scc96520@ucmo.edu CS 5600 Advanced Database Systems — Spring 2026 CRN: 23405 | Group 2

Database Description and Business Rules

The system tracks and analyzes allergy-related symptoms in pets over time. The primary users are pet owners managing pets with chronic or recurring allergies. The system allows users to create pet profiles, log daily allergy symptoms, record treatments, capture environmental conditions, and interact with the data through a natural-language chatbot powered by a Large Language Model (LLM).

Users. Each user has a unique username, a unique email address, and a timestamp recording when the account was created. A user may own one or more pets. If a user account is deleted, all associated pets and their data are also removed.

Pets. Each pet belongs to exactly one user. For each pet we store a name, species (e.g., Dog, Cat), an optional breed, and an optional date of birth. A pet has a creation timestamp. Each pet may have many symptom logs, treatment logs, and environmental factor logs associated with it. If a pet is deleted, all of its logs are also removed.

Symptom Types. The system maintains a catalog of recognized symptom categories. Each symptom type has a unique name and an optional description. Examples include Itching, Skin Redness, Sneezing, Ear Inflammation, Paw Licking, Watery Eyes, and Nasal Discharge. Symptom types are referenced by symptom logs and cannot be deleted while logs reference them.

Symptom Logs. Each symptom log records a single observed symptom occurrence for a specific pet. A symptom log references exactly one pet and exactly one symptom type. Each log includes a severity rating constrained to a scale of 1 (mild) to 10 (severe), optional free-text notes describing the observation, and a timestamp indicating when the symptom was observed. The primary query pattern is retrieving symptoms for a given pet within a date range.

Treatments. The system maintains a catalog of available treatments. Each treatment has a unique name, a treatment type that must be one of: medication, topical, dietary, or therapy, and an optional description. Examples include Apoquel (medication), Cytopoint Injection (medication), Medicated Shampoo (topical), Grain-Free Diet (dietary), Hydrocortisone Cream (topical), and Allergy Immunotherapy (therapy). Treatments cannot be deleted while treatment logs reference them, preserving historical accuracy.

Treatment Logs. Each treatment log records a single treatment administration for a specific pet. A treatment log references exactly one pet and exactly one treatment. Each log includes an optional dosage description, optional notes, and a timestamp indicating when the treatment was administered. Treatment logs allow correlation between treatment timing and symptom severity changes.

Environmental Factor Types. The system maintains a catalog of environmental factor categories. Each factor type has a unique name, an optional unit of measurement (e.g., index, °F, %), and an optional description.

Examples include Pollen Count (index on a 0–12 scale), Temperature (°F), Humidity (%), and Mold Spore Count (index on a 0–12 scale).

Environmental Factor Logs. Each environmental factor log records an environmental condition reading associated with a specific pet's location. A log references exactly one pet and exactly one environmental factor type. Each log includes a numeric value (up to two decimal places of precision), optional notes, and a timestamp. Environmental factor logs enable correlation between environmental conditions and symptom flare-ups (e.g., pollen counts above 8.0 correlating with increased symptom severity).

LLM Query Logs. Every natural-language question submitted to the chatbot is logged for auditing and debugging purposes. Each log entry references the user who asked the question and stores the original natural-language query, the SQL generated by the LLM, the response summary returned to the user, a success flag indicating whether the query executed correctly, an optional error message if the query failed, and a creation timestamp.

Business Rules Summary

1. A user must have a unique username and a unique email address.
2. Each pet belongs to exactly one user; a user may own many pets.
3. Deleting a user cascades deletion to all their pets and associated logs.
4. Deleting a pet cascades deletion to all its symptom logs, treatment logs, and environmental factor logs.
5. Each symptom type name must be unique across the system.
6. Symptom severity must be an integer between 1 and 10, inclusive.
7. Each treatment name must be unique; treatment type must be one of: medication, topical, dietary, or therapy.
8. Treatments cannot be removed from the catalog while treatment logs reference them.
9. Environmental factor values are stored as numeric with up to two decimal places of precision.
10. Every LLM chatbot interaction is logged with the original query, generated SQL, response, and success status.

NoSQL Database Model — Document Database

Why Document Model

The Document Database Model (e.g., MongoDB) is a natural fit for this system because the data is inherently pet-centric. All logging data — symptoms, treatments, and environmental factors — revolves around a specific pet owned by a specific user. A document store allows us to embed or reference related data within pet-centric documents, reducing the need for joins and enabling efficient retrieval of a pet's complete allergy history in a single query.

Collections

The system is organized into the following collections:

Collection 1 — users

Contains user account information and an embedded array of their pets.

Fields:

- `_id` (ObjectId)
- `username` (string, unique)
- `email` (string, unique)
- `created_at` (date)
- `pets` (array of embedded pet documents)
 - `pet_id` (string)
 - `name` (string)
 - `species` (string)
 - `breed` (string, optional)
 - `date_of_birth` (date, optional)
 - `created_at` (date)

Collection 2 — `symptom_types`

Reference collection for recognized symptom categories.

Fields:

- `_id` (ObjectId)
- `name` (string, unique)
- `description` (string, optional)

Collection 3 — `treatments`

Reference collection for available treatments.

Fields:

- `_id` (ObjectId)
- `name` (string, unique)
- `treatment_type` (string — one of: medication, topical, dietary, therapy)
- `description` (string, optional)

Collection 4 — `env_factor_types`

Reference collection for environmental factor categories.

Fields:

- `_id` (ObjectId)
- `name` (string, unique)
- `unit` (string, optional)
- `description` (string, optional)

Collection 5 — `symptom_logs`

Records each observed symptom occurrence for a pet.

Fields:

- `_id` (ObjectId)
- `pet_id` (string, references `users.pets.pet_id`)

- symptom_type (string — denormalized symptom type name)
- severity (integer, 1–10)
- notes (string, optional)
- logged_at (date)

Collection 6 — treatment_logs

Records each treatment administration for a pet.

Fields:

- _id (ObjectId)
- pet_id (string, references users.pets.pet_id)
- treatment_name (string — denormalized treatment name)
- treatment_type (string)
- dosage (string, optional)
- notes (string, optional)
- administered_at (date)

Collection 7 — env_factor_logs

Records environmental condition readings for a pet's location.

Fields:

- _id (ObjectId)
- pet_id (string, references users.pets.pet_id)
- factor_name (string — denormalized factor type name)
- unit (string, optional)
- value (number)
- notes (string, optional)
- logged_at (date)

Collection 8 — llm_query_logs

Audit log for every natural-language chatbot interaction.

Fields:

- _id (ObjectId)
- user_id (string, references users._id)
- natural_language_query (string)
- generated_sql (string, optional)
- response_summary (string, optional)
- success (boolean)
- error_message (string, optional)
- created_at (date)

Document Data Model Diagram

MongoDB Database pet_allergy_tracker			
users	symptom_types	treatments	env_factor_types
_id	_id	_id	_id
username	name	name	name
email	description	treatment_type	unit
created_at		description	description
pets[			
pet_id			
name			
species			
breed			
dob			
]			
symptom_logs	treatment_logs	env_factor_logs	llm_query_logs
_id	_id	_id	_id
pet_id	pet_id	pet_id	user_id
symptom_type	treatment_name	factor_name	nl_query
severity	treatment_type	unit	gen_sql
notes	dosage	value	response
logged_at	notes	notes	success
	administered_at	logged_at	error_msg
			created_at

Relationships:

users.pets[].pet_id —< symptom_logs.pet_id

users.pets[].pet_id —< treatment_logs.pet_id

users.pets[].pet_id —< env_factor_logs.pet_id

users._id —< llm_query_logs.user_id

Sample Data

Collection: users

```
{
  "_id": ObjectId("68001a1b2c3d4e5f60708090"),
  "username": "sam",
  "email": "sam@example.com",
  "created_at": ISODate("2025-12-15T00:00:00Z"),
  "pets": [
    {
      "pet_id": "pet_001",
```

```

    "name": "Finn",
    "species": "Dog",
    "breed": "Golden Retriever",
    "date_of_birth": ISODate("2021-03-15T00:00:00Z"),
    "created_at": ISODate("2025-12-15T00:00:00Z")
  }
]
}

```

Collection: symptom_types

```

{ "_id": ObjectId("68002a1b2c3d4e5f60710001"), "name": "Itching", "description":
  "Generalized scratching and itching behavior" }
{ "_id": ObjectId("68002a1b2c3d4e5f60710002"), "name": "Skin Redness",
  "description": "Visible redness or inflammation on the skin" }
{ "_id": ObjectId("68002a1b2c3d4e5f60710003"), "name": "Sneezing", "description":
  "Repeated sneezing episodes" }
{ "_id": ObjectId("68002a1b2c3d4e5f60710004"), "name": "Ear Inflammation",
  "description": "Redness, swelling, or discharge inside the ears" }
{ "_id": ObjectId("68002a1b2c3d4e5f60710005"), "name": "Paw Licking",
  "description": "Excessive licking or chewing of paws" }
{ "_id": ObjectId("68002a1b2c3d4e5f60710006"), "name": "Watery Eyes",
  "description": "Excessive tearing or eye discharge" }
{ "_id": ObjectId("68002a1b2c3d4e5f60710007"), "name": "Nasal Discharge",
  "description": "Runny nose or nasal drip" }

```

Collection: treatments

```

{ "_id": ObjectId("68003a1b2c3d4e5f60720001"), "name": "Apoquel",
  "treatment_type": "medication", "description": "Oclacitinib – daily oral JAK
  inhibitor for allergic itch" }
{ "_id": ObjectId("68003a1b2c3d4e5f60720002"), "name": "Cytopoint Injection",
  "treatment_type": "medication", "description": "Lokivetmab – monthly injectable
  antibody targeting IL-31" }
{ "_id": ObjectId("68003a1b2c3d4e5f60720003"), "name": "Medicated Shampoo",
  "treatment_type": "topical", "description": "Chlorhexidine/ketoconazole shampoo
  for skin infections" }
{ "_id": ObjectId("68003a1b2c3d4e5f60720004"), "name": "Grain-Free Diet",
  "treatment_type": "dietary", "description": "Limited-ingredient grain-free kibble
  to reduce food allergens" }
{ "_id": ObjectId("68003a1b2c3d4e5f60720005"), "name": "Hydrocortisone Cream",
  "treatment_type": "topical", "description": "OTC 1% hydrocortisone for localized
  hot spots" }
{ "_id": ObjectId("68003a1b2c3d4e5f60720006"), "name": "Allergy Immunotherapy",
  "treatment_type": "therapy", "description": "Subcutaneous allergy shots based on
  intradermal testing" }

```

Collection: env_factor_types

```
{ "_id": ObjectId("68004a1b2c3d4e5f60730001"), "name": "Pollen Count", "unit":
"index", "description": "Daily pollen index (0-12 scale)" }
{ "_id": ObjectId("68004a1b2c3d4e5f60730002"), "name": "Temperature", "unit":
"°F", "description": "Outdoor temperature in Fahrenheit" }
{ "_id": ObjectId("68004a1b2c3d4e5f60730003"), "name": "Humidity", "unit": "%",
"description": "Relative humidity percentage" }
{ "_id": ObjectId("68004a1b2c3d4e5f60730004"), "name": "Mold Spore Count", "unit":
"index", "description": "Daily mold spore index (0-12 scale)" }
```

Collection: symptom_logs

```
{
  "_id": ObjectId("68005a1b2c3d4e5f60740001"),
  "pet_id": "pet_001",
  "symptom_type": "Itching",
  "severity": 3,
  "notes": "Mild scratching after walk",
  "logged_at": ISODate("2025-12-15T14:30:00Z")
}
{
  "_id": ObjectId("68005a1b2c3d4e5f60740002"),
  "pet_id": "pet_001",
  "symptom_type": "Paw Licking",
  "severity": 2,
  "notes": "Brief paw licking after breakfast",
  "logged_at": ISODate("2025-12-15T18:00:00Z")
}
{
  "_id": ObjectId("68005a1b2c3d4e5f60740003"),
  "pet_id": "pet_001",
  "symptom_type": "Itching",
  "severity": 8,
  "notes": "Worst itching day – scratching nonstop",
  "logged_at": ISODate("2025-12-24T14:00:00Z")
}
{
  "_id": ObjectId("68005a1b2c3d4e5f60740004"),
  "pet_id": "pet_001",
  "symptom_type": "Skin Redness",
  "severity": 7,
  "notes": "Hot spots forming on flanks",
  "logged_at": ISODate("2025-12-25T18:00:00Z")
}
{
  "_id": ObjectId("68005a1b2c3d4e5f60740005"),
  "pet_id": "pet_001",
  "symptom_type": "Itching",
  "severity": 2,
  "notes": "Much better after Cytopoint injection",
```

```
"logged_at": ISODate("2025-12-29T14:00:00Z")
}
```

Collection: treatment_logs

```
{
  "_id": ObjectId("68006a1b2c3d4e5f60750001"),
  "pet_id": "pet_001",
  "treatment_name": "Apoquel",
  "treatment_type": "medication",
  "dosage": "16mg tablet",
  "notes": "Morning dose with food",
  "administered_at": ISODate("2025-12-15T13:00:00Z")
}
{
  "_id": ObjectId("68006a1b2c3d4e5f60750002"),
  "pet_id": "pet_001",
  "treatment_name": "Cytopoint Injection",
  "treatment_type": "medication",
  "dosage": "40mg injection",
  "notes": "Monthly Cytopoint – administered at vet",
  "administered_at": ISODate("2025-12-28T20:00:00Z")
}
{
  "_id": ObjectId("68006a1b2c3d4e5f60750003"),
  "pet_id": "pet_001",
  "treatment_name": "Medicated Shampoo",
  "treatment_type": "topical",
  "dosage": "Full body wash",
  "notes": "Weekly bath – 10 min contact time",
  "administered_at": ISODate("2025-12-15T23:00:00Z")
}
{
  "_id": ObjectId("68006a1b2c3d4e5f60750004"),
  "pet_id": "pet_001",
  "treatment_name": "Hydrocortisone Cream",
  "treatment_type": "topical",
  "dosage": "Thin layer on affected area",
  "notes": "Applied to hot spots on belly",
  "administered_at": ISODate("2025-12-25T02:00:00Z")
}
{
  "_id": ObjectId("68006a1b2c3d4e5f60750005"),
  "pet_id": "pet_001",
  "treatment_name": "Grain-Free Diet",
  "treatment_type": "dietary",
  "dosage": "2 cups daily",
  "notes": "Started grain-free diet",
  "administered_at": ISODate("2025-12-15T13:30:00Z")
}
```

Collection: env_factor_logs

```
{
  "_id": ObjectId("68007a1b2c3d4e5f60760001"),
  "pet_id": "pet_001",
  "factor_name": "Pollen Count",
  "unit": "index",
  "value": 3.2,
  "notes": "Low pollen day",
  "logged_at": ISODate("2025-12-15T12:00:00Z")
}
{
  "_id": ObjectId("68007a1b2c3d4e5f60760002"),
  "pet_id": "pet_001",
  "factor_name": "Temperature",
  "unit": "°F",
  "value": 45.0,
  "notes": null,
  "logged_at": ISODate("2025-12-15T12:00:00Z")
}
{
  "_id": ObjectId("68007a1b2c3d4e5f60760003"),
  "pet_id": "pet_001",
  "factor_name": "Pollen Count",
  "unit": "index",
  "value": 10.2,
  "notes": "Peak pollen – extreme",
  "logged_at": ISODate("2025-12-24T12:00:00Z")
}
{
  "_id": ObjectId("68007a1b2c3d4e5f60760004"),
  "pet_id": "pet_001",
  "factor_name": "Humidity",
  "unit": "%",
  "value": 70.0,
  "notes": null,
  "logged_at": ISODate("2025-12-23T12:00:00Z")
}
{
  "_id": ObjectId("68007a1b2c3d4e5f60760005"),
  "pet_id": "pet_001",
  "factor_name": "Mold Spore Count",
  "unit": "index",
  "value": 5.8,
  "notes": "Mold also elevated",
  "logged_at": ISODate("2025-12-24T12:00:00Z")
}
```

Collection: llm_query_logs

```
{
  "_id": ObjectId("68008a1b2c3d4e5f60770001"),
  "user_id": "68001a1b2c3d4e5f60708090",
  "natural_language_query": "What allergy symptoms has Finn had in the last 30 days?",
  "generated_sql": "SELECT st.name, sl.severity, sl.notes, sl.logged_at FROM symptom_logs sl JOIN symptom_types st ON sl.symptom_type_id = st.symptom_type_id WHERE sl.pet_id = 1 AND sl.logged_at >= NOW() - INTERVAL '30 days' ORDER BY sl.logged_at DESC",
  "response_summary": "In the last 30 days, Finn experienced 15 symptom occurrences. The most frequent symptom was Itching (9 times, avg severity 4.2). Paw Licking appeared 3 times. Severity peaked at 7 on Feb 3-4 during a pollen spike.",
  "success": true,
  "error_message": null,
  "created_at": ISODate("2026-02-13T20:00:00Z")
}
{
  "_id": ObjectId("68008a1b2c3d4e5f60770002"),
  "user_id": "68001a1b2c3d4e5f60708090",
  "natural_language_query": "Has Finn's itching gotten worse this week?",
  "generated_sql": "SELECT sl.severity, sl.logged_at FROM symptom_logs sl JOIN symptom_types st ON sl.symptom_type_id = st.symptom_type_id WHERE sl.pet_id = 1 AND st.name = 'Itching' AND sl.logged_at >= NOW() - INTERVAL '7 days' ORDER BY sl.logged_at",
  "response_summary": "Finn's itching has improved this week. His severity dropped from 4 on Feb 6 to a steady 2 over the last several days, indicating the Cytopoint injection is working well.",
  "success": true,
  "error_message": null,
  "created_at": ISODate("2026-02-13T20:05:00Z")
}
```